



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 04

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Program:

```
def fibonacci_iterative(n):  
    series = []  
    a, b = 0, 1  
    for i in range(n):  
        series.append(a)  
        a, b = b, a + b  
    return series  
  
def fibonacci_recursive(n):  
    if n <= 1:  
        return n  
    return fibonacci_recursive(n - 1) + fibonacci_recursive(n - 2)  
  
n = 6  
series = []  
for i in range(n):  
    series.append(fibonacci_recursive(i))  
print("Iterative:", fibonacci_iterative(n))  
print("Recursive:", series)
```

Output:

main.py	Visualize	Run ▶	Output
<pre>1 def fibonacci_iterative(n): 2 series = [] 3 a, b = 0, 1 4 5 for i in range(n): 6 series.append(a) 7 a, b = b, a + b 8 9 return series 10 11 12 def fibonacci_recursive(n): 13 if n <= 1: 14 return n 15 return fibonacci_recursive(n - 1) + fibonacci_recursive(n - 2)</pre>			<pre>Iterative: [0, 1, 1, 2, 3, 5] Recursive: [0, 1, 1, 2, 3, 5]</pre>